

TO-DO LIST APPLICATION

DETAILED ABSTRACT

This project is a simple **To-Do List application** created using **Python** and the **Tkinter** library. Tkinter is used to make a small window where users can interact with buttons and text fields. The main goal of this project is to help users manage their daily tasks in an easy way.

When the program starts, a window appears with an input box, an **Add Task** button, a **Delete Task** button, and a list area. The user can type a task in the input box and click the **Add Task** button. If the user clicks the button without typing anything, a warning message will appear asking them to enter a task. When a valid task is entered, it is saved into a list inside the program and also displayed in the list area.

To remove a task, the user must select the task from the list and click the **Delete Task** button. If the user tries to delete without selecting any task, a warning message appears asking them to select a task first. The selected task is then removed from both the list on the screen and the list stored inside the program.

This project uses basic programming concepts such as functions, lists, conditional statements, and event-driven programming. It is a beginner-friendly project that shows how to create a simple graphical user interface and perform basic task operations. Even though the tasks are not saved permanently in a file or database, the application still helps users organize their tasks during the time the program is running.

Overall, this To-Do List application is easy to use and provides a simple way to manage daily tasks, making it useful for students and beginners who want to learn GUI programming in Python.

Test Case	Input	Expected Output	Result
Add Task with valid text	"Buy groceries"	Task appears in list	Pass
Add Task with empty text	" " (blank)	Warning message appears	Pass
Delete task after selecting	Select task → Click Delete	Task removed from list	Pass
Delete without selecting	Click Delete without selection	Warning message appears	Pass

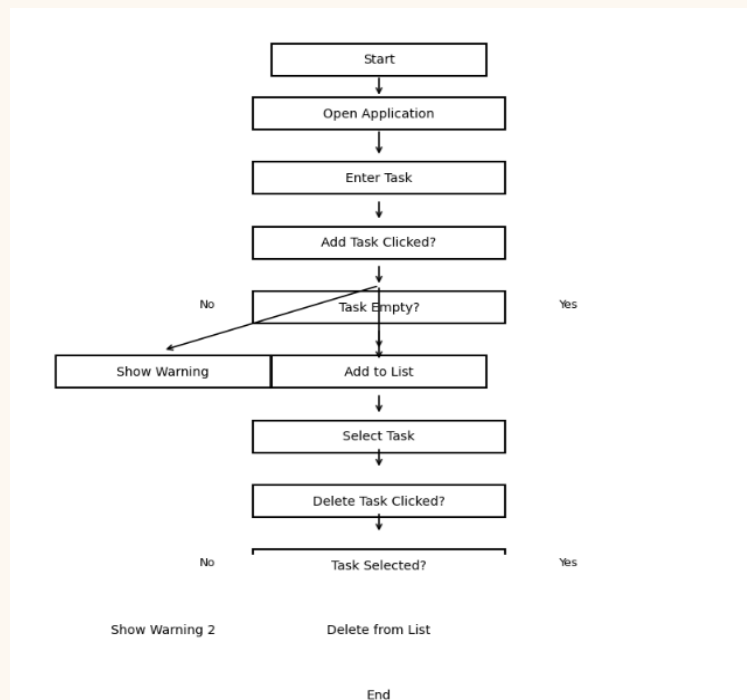
ALGORITHM

Algorithm for Adding Task

1. Start
2. Read task from input box
3. If task is empty
 - Show warning message
 - Go back to step 2
4. Else
 - Add task to the list
 - Show task in the listbox
 - Clear the input box
5. End

Algorithm for Deleting Task

1. Start
2. Check if a task is selected from the list
3. If no task is selected
 - Show warning message
 - Go back to step 2
4. Else
 - Remove selected task from the list
 - Remove selected task from the listbox
5. End



The application successfully meets its objectives by providing a clean interface and proper validation messages. Although it does not store tasks permanently, it serves as a strong foundation for learning GUI development in Python. In future, the project can be enhanced by adding features like saving tasks to a file or database, setting deadlines, and adding priority levels.